



Quantum Radar

Design a radar system that has the following

- The radar handles info sent from the physical radar
 - Plate number, Date, Car type (e.g., Private, Truck, Bus), Speed, Seatbelt status (fastened / not)
- a set of defined rules, like
 - Truck speed shouldn't exceed 60
 - Private car speed shouldn't exceed 80
 - seat belt should be fastened
- on each observation, you may generate zero or more violations according to the available rules
- if there are violations a fine should be issued that has information about the violations and the fees of each, printing exactly:

Traffic fine for car ABC1234

Total amount: 400 EGP

Violations:

- Seatbelt not fastned : 100 EGP

- speed of 94 exceeded max allowed 80 : 300 EGP

- get all fines: plate number with total amount
- get all violated rules with count for each
- The system should be designed to be extensible, allowing flexibility for adding rules without modifying the Radar

Notes

1. Create a Main class to demonstrate the operation of your solution
2. Stick to simple design
3. Assume what is needed for the solution to work
4. No need to think about the physical radar
5. Don't copy any content from this file, Don't take screenshots, Don't copy the file into your workspace or use it for any purpose
6. Don't use AI